

N-gram Language Model Report

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Part 1: Results

Model	Train PPL	Dev PPL	Test PPL
Unigram	2083.65	1907.91	1923.24
Bigram	180.44	17679.66	17691.82
Trigram	7.87	4.02e7	3.90e7

Using maximum likelihood estimation (MLE), I developed unigram, bigram, and trigram models, assessed perplexity on the train, development, and test sets, and carried out preliminary behavioral analysis. Without the use of NLP packages, all models were fully implemented in Python.

This submission includes a preliminary discussion for the other portions of the assignment as well as the complete implementation and outcomes for Part 1. In the resubmission, I intend to expand on the OPT-1.3B language-model analysis and interpolation experiments.

2. Preparing Data

Each line in the dataset consists of whitespace-tokenized sentences. I carried out the following preprocessing actions:

Tokenization

The data is already whitespace-tokenized, so I used `str.split()`.

Handling Rare Words

Tokens with frequency < 3 in the training split were mapped to $\langle \text{UNK} \rangle$. The final vocabulary size was 26,602 types.

Sentence Boundary Markers

Two $\langle \text{START} \rangle$ tokens were added at the beginning and one $\langle \text{STOP} \rangle$ token at the end of each sentence.

Dev/Test Handling

Tokens unseen in training were mapped to $\langle \text{UNK} \rangle$ during evaluation.

3. MLE N-gram Language Models

3.1 Parameter Estimation

$$P_{\text{uni}}(w) = \frac{\text{count}(w)}{\text{total training tokens}}$$

$$P_{\text{bi}}(w \mid w') = \frac{\text{count}(w', w)}{\text{count}(w')}$$

$$P_{\text{tri}}(w \mid w_{i-2}, w_{i-1}) = \frac{\text{count}(w_{i-2}, w_{i-1}, w)}{\text{count}(w_{i-2}, w_{i-1})}$$

Following the assignment rule, the token after $\langle \text{START} \rangle$, $\langle \text{START} \rangle$ uses the bigram probability $P(w \mid \langle \text{START} \rangle)$ even in the trigram model.

3.2 Perplexity

$$\text{PPL} = \exp \left(-\frac{1}{M} \sum_{i=1}^M \log P(w_i \mid \cdot) \right)$$

A floor value of 10^{-12} was used to avoid zero-probability log issues.

5. Planned Work for Parts 2 and 3

For the resubmission, I will complete:

- Linear interpolation experiments
- OPT-1.3B analysis